Beyond the Hyper-Scalers

INDEPENDENT THEMATIC RESEARCH

EQUITY STRATEGY · PORTFOLIO INITIATION

Quantifying the Physical Bottlenecks of the Global AI CapEx Supercycle

Overweight Energy, hold Compute as a counterweight, and size Ballast to absorb drawdown. 18 positions across six exchanges express a rotation out of hyperscaler beta and into the physical infrastructure that gates it.

• **The consensus AI trade is crowded.** Hyperscalers and their primary silicon suppliers are the obvious way to own the buildout, and the most efficiently priced. This book is built to express a different, less crowded layer of the same trade.

• **The binding constraint is shifting from silicon to substrate.** Grid interconnect queues and transformer lead times, not chip supply, increasingly gate how fast new capacity can actually come online.

• **Construction is top-down, not stock-by-stock.** Target weights are set at the bucket level (Energy 38% / Compute 32% / Ballast 30%); positions inside each bucket are held close to equal-weight so no single stock call carries the thesis.

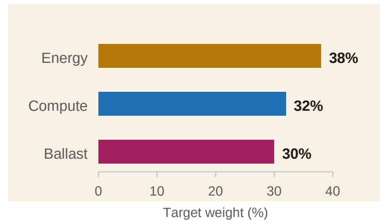
• **Compute is retained on purpose, not shrunk to a token hedge.** A portfolio that only wins if hyperscalers underperform their own suppliers is a weaker construction than one that wins on the rotation but does not collapse if the rotation stalls.

• **Every position carries a standing falsification note:** the specific condition under which its inclusion in the thesis is wrong. Two classification calls (the neocloud names, the critical-minerals ETF) are flagged as stretches rather than smoothed over. See Section 4.

KEY METRICS

Index level	**151.89**
YTD (USD)	**+32.9%**
1-year (USD)	**+51.9%**
Volatility (ann.)	**25.6%**
Max drawdown	**–14.3%**

BUCKET ALLOCATION



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2. PORTFOLIO CONSTRUCTION

Bucket Weights and Factor Logic

The book is organized into three thematic buckets rather than picked stock-by-stock. Target weights are set at the bucket level; positions within a bucket are then held close to equal-weight, so no single stock-picking call inside a bucket is doing outsized work. The bet is on the theme, not on any individual name being the best operator in it.

Bucket	Weight	N	Structural exposure
Energy	38.0%	7	Baseload generation, transmission, fuel supply, land and interconnect rights: the most direct expression of the thesis
Compute	32.0%	7	The hardware layer itself: a macro counterweight against being wrong about the timing of the rotation
Ballast	30.0%	4	Cross-asset hedges, monetary safety assets, and development-market value, sized to absorb drawdown

Energy is weighted above Compute deliberately: it is the more direct, less crowded expression of the thesis. Compute is kept at close to a third of the book on purpose: a portfolio that only wins if hyperscalers underperform their own suppliers is a weaker construction than one that wins on the rotation but does not collapse if the rotation stalls. Ballast is sized to absorb drawdown from the other two buckets rather than to add a competing return thesis.

Valuation Discipline and Entry Criteria

A weight target says how much to hold, not what price makes it worth holding. Each bucket carries an explicit initiation threshold and a trim rule, so a position that re-rates away from its thesis is cut on a stated condition rather than on sentiment.

Bucket	Initiation threshold and trim discipline
Energy	Initiation threshold: Forward EV/EBITDA < 14x or FCF Yield > 5.0%. Position trimmed if NTM EV/EBITDA expands beyond 20x without corresponding reserve/interconnect capacity growth.
Compute	Cap total exposure if Basket Forward P/E > 32x or NTM PEG > 1.8. Serves as pure valuation-disciplined macro insurance.
Ballast	Rebalance when yield spread / discount to NAV deviates by >1.5 standard deviations from the 3-year trailing mean.

3. MICROECONOMIC MAPPING

Position-Level Rationale

Energy & Power (38.0%, 7 positions)

- **URA: Global X Uranium ETF · ~5.4%**
  Nuclear is the fastest-permitting large-scale answer to hyperscaler 24/7 power demand; the broadest, most liquid way to hold exposure to the fuel cycle.
- **NLR: VanEck Uranium & Nuclear ETF · ~5.4%**
  Deliberate overlap with URA on the nuclear theme rather than diversifying away from it. Conviction is high enough to accept concentration in exchange for broader coverage across the fuel cycle and regulated utilities.

- **PETR4.SA: Petrobras · ~5.4%**
  A large, cash-generative conventional producer as a hedge against nuclear permitting delays: thermal generation remains the marginal supplier during interim grid deficits.
- **2222.SR: Saudi Aramco · ~5.4%**
  Same logic as PETR4.SA with a different sovereign and cost-curve position; diversifies the conventional-energy leg across geography rather than concentrating it in one producer.
- **LB: LandBridge Company · ~5.4%**
  Surface and water rights exposure in the Permian Basin: a direct read on interconnection-queue monetization and data-centre land-siting, the physical layer underneath the generation names.
- **NBIS: Nebius Group · ~5.4%**
  Classified here as a power consumer, which is a stretch I flag rather than hide: Nebius is a compute landlord: capex-funded, customer-concentrated, and competing with the hyperscalers it sells to. It sits in Energy because its unit economics are gated by power access, not because it generates power.
- **CRWV: CoreWeave · ~5.4%**
  Same classification stretch as NBIS, and the two overlap enough operationally that this is closer to one position expressed twice than two independent bets (see Section 4).

Compute & Hardware (32.0%, 7 positions)

- **TSM: Taiwan Semiconductor Manufacturing · ~4.6%**
  The central foundry bottleneck and the counterweight the thesis has to survive: near-monopoly on advanced-node and CoWoS packaging capacity captures the demand growth directly if the rotation thesis is wrong or early.
- **AMD: Advanced Micro Devices · ~4.6%**
  Second-source accelerator exposure, kept smaller than a conviction bet would be: this bucket is insurance against the thesis, not a parallel thesis of its own.
- **PLTR: Palantir Technologies · ~4.6%**
  Software/deployment-layer exposure, included so the bucket is not pure hardware and does not miss the demand side entirely.
- **VGT: Vanguard Information Technology ETF · ~4.6%**
  A broad, low-conviction sleeve that keeps the bucket from being a handful of single-name bets on a theme this book is intentionally underweighting.
- **2357.TW: ASUSTeK Computer · ~4.6%**
  Server assembly exposure: margin-thin and customer-concentrated, and the legacy PC cycle still outweighs the AI server line in reported revenue. Held as a read on physical assembly capacity, not a pure AI-server bet.
- **^KS11: KOSPI Composite Index · ~4.6%**
  An index proxy for Korea's memory manufacturing exposure to the buildout, avoiding having to pick between the two dominant memory makers.
- **AINF.L: AI Infrastructure UCITS ETF (LSE) · ~4.6%**
  A packaged, diversified sleeve of infrastructure and connectivity names as a lower-maintenance complement to the single-name bets elsewhere in the bucket.

Macro Ballast (30.0%, 4 positions)

- **GLD: SPDR Gold Shares · ~7.5%**
  Classic drawdown hedge, uncorrelated to the CapEx cycle either way. Held to reduce portfolio volatility, not to express a view.
- **AVDV: Avantis Intl Small Cap Value ETF · ~7.5%**
  Deliberately away from the AI theme entirely: developed-market, value-tilted, small-cap exposure as a genuine diversifier rather than a disguised extra bet on the same trade.

- **JPM: JPMorgan Chase & Co. · ~7.5%**
  A quality financial with real exposure to the debt financing the CapEx cycle runs on, but broad enough not to live or die on it.
- **RARA11.SA: Rare Earths / Critical Minerals ETF · ~7.5%**
  The other classification stretch I flag openly: a concentrated, policy-driven basket that tracks the same technology-materials cycle as the rest of the book, so in most states it does less true ballasting than its bucket label implies. The exception is an export-control shock, where it is the one line in the book that re-rates upward. It also traded in only about 8% of the measured window, which limits how much its long-horizon return figures should be trusted.

4. ANALYTICAL DISCLOSURES

Known Portfolio Weaknesses

Every position in the live dashboard carries a standing falsification note: the specific condition under which its place in the thesis is wrong, refreshed alongside the daily data pull. The intent is to make it structurally harder to only look for evidence that confirms the thesis. The open weaknesses, stated plainly:

Neocloud classification overlap. NBIS and CRWV are functionally compute-infrastructure lessors, not power generators, and their high operational correlation compresses the 18-position book closer to 17 independent bets.

Ballast misclassification and liquidity drag. RARA11.SA is assigned to Ballast but is correlated with the same technology-materials cycle the rest of the book is long, and its thin trading history (~8% active coverage in the measured window) adds noise to any long-horizon variance figure.

Temporal horizon mismatch. Daily deviation tracking compares short-term market noise against a multi-year infrastructure lead-time thesis (permitting and transformer procurement run three to five years). Daily relative performance between Energy and Compute is an operational monitoring routine, not statistical validation.

Near-Term Monitoring Catalysts

The mismatch above is real but not unanswerable. Three things resolve inside nought to four quarters and bear directly on whether the physical constraint is binding, which is what daily price action cannot tell you:

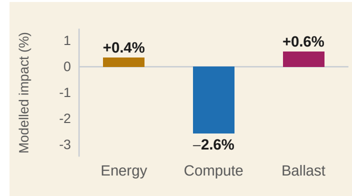
- **Policy & Grid Reform.** Track compliance timelines and cluster-study clearing under FERC Order 2023 (US interconnection queue reform), alongside PJM and ERCOT capacity auction clearing prices. Queue throughput is the single most direct read on whether the interconnect bottleneck is binding or easing.
- **Supply Chain Lead Times.** Monitor quarterly management guidance on high-voltage transformer and switchgear lead times from key equipment OEMs — Eaton Corp (ETN), Hubbell (HUBB) and Schneider Electric. Guidance revisions front-run the physical constraint by several quarters.
- **Nuclear PPAs & Co-location.** Track announcements of direct-connect nuclear Power Purchase Agreements (Constellation Energy, Talen Energy, Vistra) and hyperscaler co-location approvals. Each signed PPA is a datable observation that baseload scarcity is being priced.

5. RISK FRAMEWORK

Scenario Stress-Testing

An empirical covariance matrix, estimated over a year of daily data across six currencies (one with three weeks of trading history), would look statistically rigorous while being almost entirely noise. The portfolio uses stated-assumption scenario shocks instead.

Interconnection queues and transformer lead times, not chip supply, become the binding constraint; announced data-centre builds slip. Modelled portfolio impact **–1.6%** (Energy +0.4%, Compute –2.6%, Ballast +0.6%). This is the scenario the book is built to profit from being right about: the one where a negative headline number still confirms the thesis is working as designed.



CapEx Retrenchment (cyclical digestion): hyperscalers cut AI capital-spending guidance and the supercycle is re-rated as an ordinary cycle. This is the scenario the Compute bucket exists to cushion against, and the one most likely to hurt every bucket at once. No point estimate is modelled for this scenario. The position sizing, not a forecast number, is the hedge.

Geopolitical Shock (Taiwan Strait escalation): advanced foundry output is interrupted and export controls tighten on strategic materials. The tail scenario the portfolio is least hedged against: TSM, 2357.TW and ^KS11 all sit in the impact zone and are closer to one bet than three. RARA11.SA is the one line that works here, since the same export controls that interrupt foundry output bid rare earths up, but at 7.5% it cannot offset a Compute bucket four times its size. Also left unmodelled rather than assigned a manufactured number.

> **Why stated assumptions instead of a fitted model.** A covariance matrix estimated from a year of daily data across six currencies would look statistically rigorous and would be almost entirely noise, given how little history several positions have. Stating one assumption directly and leaving the other two scenarios qualitative is less impressive-looking than assigning every scenario a precise number, and it is more honest about what the data can actually support.

6. TECHNICAL STACK

Ingestion Pipeline & Architecture

The dashboard is built and deployed on Streamlit. Text generation for the daily note and the per-position falsification commentary runs through a deterministic, rule-based renderer rather than a live LLM call. No API key is configured in the deployed version, so the analytical judgment embedded in the classifications and the self-critique is fixed at build time, not generated fresh each day. What refreshes on weekdays is the price, volume, and news data feeding that fixed framework: Google News RSS, SEC EDGAR's Atom feed for filings, and the Yahoo Finance chart v8 endpoint for prices. This is a deliberate choice worth stating plainly, since it affects what the analytical work in the project actually is: it sits in the research and portfolio-construction decisions, not in a model narrating markets live.